It appears from the table that as x gets close to $x = 2, f(x) = x^2 + 3$ gets close to 7. This is not surprising since if we now calculate $f(x)$ at $x = 2$, we obtain $f(2) = 2^2 + 3 = 7$.

In order to guess at this limit, we didn't have to evaluate $x^2 + 3$ at $x = 2$.

That is, as x approaches 2 from either the left (values lower than 2) or right (values higher than 2) the functional values $f(x)$ are approaching 7 from either side; that is, when x is near 2, $f(x)$ is near 7. The above situation is described in a condensed form:

The value 7 is the left limit of $f(x)$ as x approaches 2 from the left as well as 7 is the right limit of $f(x)$ as x approaches 2 from the right and write :

$f(x) \to 7$ as $x \to 2^-$ and $f(x) \to 7$ as $x \to 2^+$

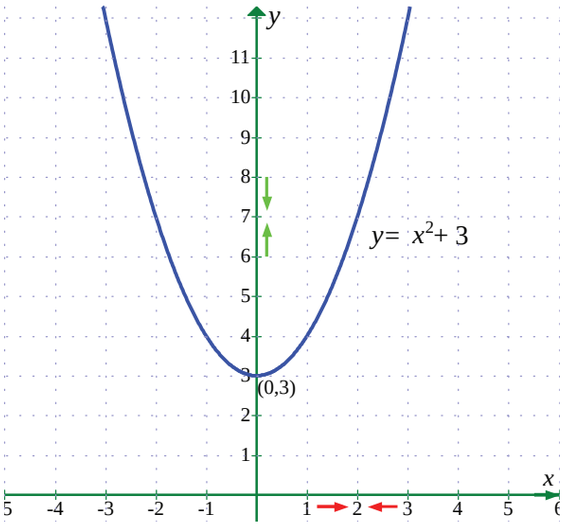


Fig. 9.1

or

$\lim_{x \to 2^-} f(x) = 7$ and $\lim_{x \to 2^+} f(x) = 7$.

Note also that $\lim_{x \to 2^-} f(x) = 7 = \lim_{x \to 2^+} f(x)$. The common value is written as $\lim_{x \to 2} f(x) = 7$.

We also observe that the limit is a definite real number. Here, definiteness means that $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$ are the same and

$\lim_{x \to 2} f(x) = \lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$ is a unique real number.

The figure in Fig. 9.1 explains the geometrical significance of the above discussion of the behaviour of $f(x) = x^2 + 3$ as $x \to 2$.

Illustration 9.2

Next, let us look at the rational function $f(x) = \dfrac{16 - x^2}{4 + x}$.

The domain of this function is $\mathbb{R} \setminus \{-4\}$. Although $f(-4)$ is not defined, nonetheless, $f(x)$ can be calculated for any value of x near -4 because the symbol $\lim_{x \to -4} \left(\dfrac{16 - x^2}{4 + x} \right)$ says that we consider values of x that are close to -4 but not equal to -4. The table below gives the values of $f(x)$ for values of x that approach -4.

$(x < -4)$ $(x \to -4^-)$	$f(x)$	$(x > -4)$ $(x \to -4^+)$	$f(x)$
-4.1	8.1	-3.9	7.9
-4.01	8.01	-3.99	7.99
-4.001	8.001	-3.999	7.999

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